

PHYSICI

Physical Science

Grade 12, Semester 1 study outline

A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.

01. Formation of Elements During Stellar Formation and Evolution

1. Stellar Nebula

A large cloud of gas and dust where stars begin to form.

2. Protostar

Gas and dust contract because of gravity. The forming star becomes hotter and denser.

3. Main Sequence Star

Longest stage of a star's life. Hydrogen is fused into helium.

4. Giant Stage

Small to medium stars expand after hydrogen is used up.

5. Final Stage

1. Why can stars continue shining for millions or billions of years?

2. Why are elements heavier than iron usually formed during supernova explosions?

3. If stars create elements, what does this suggest about the origin of matter on Earth?

"We are made from elements formed inside stars."

Key terms

Elements heavier than iron - formed during this explosion through neutron capture

02. Polarity of Molecules

1. Electron pairs repel each other

2. Electron pairs stay as far apart as possible

3. Molecular shape depends on bond pairs and lone pairs

4. Lone pairs repel more strongly than bond pairs

1. Draw the Lewis structure

2. Count bond pairs and lone pairs

3. Use VSEPR geometry

4. Predict molecular shape and bond angles

1. Why does water (H₂O) become polar while nitrogen gas (N₂) is nonpolar?

2. How do lone pairs affect molecular shape?

Key terms

Electronegativity - ability of an atom to attract shared electrons in a chemical bond

03. Intermolecular Forces of Attraction

1. London Dispersion Forces (LDF)

2. Dipole-Dipole Forces

- Positive end of one molecule attracts the negative end of another

3. Hydrogen Bonding

1. Hydrogen bonding strongest

2. Polar molecules stronger dipole forces

3. Larger molecules stronger London forces

1. Why does water have a higher boiling point than HS?
2. Why do oil and water not mix easily?
3. Why are larger molecules more likely to have stronger London dispersion forces?

"The stronger the intermolecular forces, the harder it is for molecules to separate."

Key terms

Molecules - not only held together internally by chemical bonds. They also attract other molecules through Hydrogen bonding - stronger than ordinary dipole-dipole forces

04. Biological Macromolecules

1. Carbohydrates

Often called saccharides, carbohydrates are the body's main source of energy because they are easily converted

2. Lipids

Lipids are commonly known as fats. They are hydrophobic, meaning they do not mix with water.

3. Proteins

Proteins are made of amino acids and are considered important structural and functional molecules in living

4. Nucleic Acids

Nucleic acids store and transfer genetic information.

2. Why can oil and water not mix easily?

3. How do proteins and nucleic acids work together inside cells?

"Biomolecules are the building materials, energy sources, and information carriers of life."

Key terms

Living organisms - made of large organic molecules called biomolecules or biological macromolecules. These

Biomolecules - large molecules made from smaller building blocks called monomers

Lipids - commonly known as fats. They are hydrophobic, meaning they do not mix with water

Proteins - made of amino acids and are considered important structural and functional molecules in living

05. Rates of Chemical Reaction

1. Proper orientation of particles

2. Enough energy during collision

Particles must collide in the correct position for bonds to break and form successfully.

1. Temperature

Increasing temperature increases particle movement.

2. Concentration

Higher concentration means more particles are present in the same space.

3. Surface Area

Crushed coal burns faster than a large chunk of coal.

4. Catalyst

A catalyst speeds up a reaction by providing an alternative pathway with lower activation energy.

1. Why do reactions usually happen faster at higher temperatures?

2. Why does powdered material react faster than large solid pieces?

3. How does lowering activation energy increase reaction rate?

"Chemical reactions happen through effective collisions between particles."

Key terms

Rusting of iron - chemical change because a new substance forms

Activation energy - minimum energy needed for a reaction to occur

Higher concentration - more particles are present in the same space

06. Limiting Reactants

1. Write the balanced chemical equation
2. Find the stoichiometric mole ratio
3. Determine molar masses
4. Convert grams to moles
5. Find the actual mole ratio
6. Compare actual and stoichiometric ratios
 - If actual ratio < stoichiometric ratio first reactant is limiting
1. Why must grams be converted to moles before comparing reactants?
2. Why does the reaction stop once the limiting reactant is used up?
3. Can two reactants both be limiting at the same time?

"The limiting reactant determines how much product can be formed."

Key terms

Once the limiting reactant - completely used up, the reaction stops even if other reactants are still available

07. Sources of Energy

1. Name the source first. Renewable ba or non-renewable?
3. Hanapin ang turbine and generator, or explain why the system does not need them
4. Give one practical benefit and one limitation or consequence. Dapat parehong meron
5. Final check: naipaliwanag mo ba ang process, or pinangalanan mo lang ang source?

Sir will not give the answers yet, commit muna before checking.

Key terms

Renewable sources - naturally replenished and can be used repeatedly. Sa madaling sabi, napapalitan ang Pulangi system - the lesson's Mindanao example

08. Active Ingredients of Cleaning Products

1. Define the job: grease ba, mineral deposit, stain, visible dirt, or germs?
2. Identify the active ingredient or ingredient family on the label
6. After use, close, store, and dispose of the product exactly as directed

Sir will not give the answers yet, commit muna before checking.

Key terms

Water - polar; many oily substances are non-polar. Kaya an appropriate organic solvent can dissolve residues that water

Disinfectants - intended for nonliving surfaces. Products used safely on skin or wounds are more accurately called

09. Ingredients in Cleaning Agents

I. 4 Types Of Cleaning Agent Ingredients

1. Functional Cleaning Agents

Actively participate in breaking down dirt, grease, and stains or whitening surfaces.

A. Sodium bicarbonate (Baking Soda)

Sodium bicarbonate (NaHCO_3), commonly known as baking soda, is a mild, natural cleaning agent. It acts as a

B. Silicates

Silicates, such as sodium silicate, are commonly used in cleaning products for their alkaline and corrosion-

C. Ammonia

Ammonia (NH_3) is a powerful, fast-evaporating cleaning agent known for cutting through grease, grime, and stains.

D. Bleaching Agents

Bleaching agents are chemicals used to whiten, disinfect, and remove stains by breaking down colored

2. Biological Helpers

Break down food, stains, proteins. Enhance stain removal using enzymes to break down biological residues.

A. Enzymes

Enzymes are biological molecules (proteins) used in cleaning products to break down specific stains and

3. Stability & Texture Additives

Improve texture, shelf life, stability. Help the product maintain a good texture, consistency, and long shelf life.

A. Polymers

Polymers in cleaning agents are large molecules made up of repeating units that help trap dirt, prevent re-

B. Processing Aids

Processing aids are ingredients added to cleaning products to enhance manufacturing, improve stability, or

C. Preservatives

Preservatives are chemicals added to cleaning products to prevent the growth of bacteria, mold, and fungi,

4. Aesthetic Additives

Added for smell/appearance. Improve user experience but do not contribute directly to cleaning.

A. Colorants

- Help users differentiate between similar-looking substances

Key terms

Bleaching agents - chemicals used to whiten, disinfect, and remove stains by breaking down colored

Enzymes - biological molecules (proteins) used in cleaning products to break down specific stains and

Polymers in cleaning agents - large molecules made up of repeating units that help trap dirt, prevent re-

Processing aids - ingredients added to cleaning products to enhance manufacturing, improve stability, or

Preservatives - chemicals added to cleaning products to prevent the growth of bacteria, mold, and fungi

Fragrances - added to cleaning products to give them a pleasant smell, mask chemical odors, and enhance

10. Ancient to Modern Astronomy

I. The Greeks and the Spherical Earth

The ancient Greeks were among the first to suggest that Earth is a sphere-long before modern technology

1. Ship's Hull Disappearing First

- As ships sailed away, the hull vanished before the sails.

2. Phases of the Moon & Lunar Eclipses

- The shadow Earth casts on the Moon during a lunar eclipse is always round.

3. Eratosthenes' Measurement of Earth's Size (c. 240 BCE)

Eratosthenes, a Greek scholar and librarian in Alexandria, made the first known accurate calculation of Earth's

II. Ancient Astronomical Phenomena

Ancient astronomers carefully observed the sky and recorded unusual patterns in the stars and planets. Their

III. Laws Of Planetary Motion

The movement of planets around the Sun puzzled astronomers for centuries. Thanks to Tycho Brahe's observations

1. Law of Ellipses

- Planets orbit the Sun in an ellipse, with the Sun at one focus.

2. Law of Equal Areas

- A line from a planet to the Sun sweeps out equal areas in equal times.

3. Law of Harmonies

- The square of a planet's orbital period is proportional to the cube of its average distance from the Sun.

11. Aristotelian and Galilean Concept of Motion

I. Nature of Motion

- force exists within an object that compels it to behave in a certain manner.

II. Vertical Motion

- the behavior of objects depends upon their composition

III. Projectile Motion

- motion of an object which is moving parallel to the ground due to an 'impetus' until it is time for it to fall

IV. Acceleration in a Vacuum

It was mentioned earlier that friction is an interference to motion. This means the presence of friction can slow

12. Propagation of Light, Reflection and Refraction

Key terms

Reflection and Refraction - fundamental behaviors of light that can be explained using both the wave and

Refraction - bending of light when it passes obliquely from one medium to another of different density or

13. Wave Properties of Light

I. Experiment showing that electrons behave like waves

Light exhibits both wave-like and particle-like properties (this is called wave-particle duality).

II. Wave Properties of Light

1. Dispersion of Light

The separation of light (or other waves) into different components based on wavelength.

2. Scattering of Light

The deflection of light (or other waves) by particles or irregularities in a medium.

A. Rayleigh Scattering

occurs when light interacts with particles that are much smaller than the wavelength of the light, such as

B. Mie Scattering

occurs when light interacts with particles that are roughly the same size as the wavelength of the light.

C. Non Selective Scattering

happens when the particles are much larger than the wavelength of light. Common particles responsible

3. Interference

The phenomenon where two or more waves overlap and combine, either amplifying (constructive

4. Diffraction

The bending and spreading of waves around obstacles or through small openings, especially when the

14. Various Light Phenomena

A. Your reflection on the concave and convex sides of a spoon looks different

Have you ever noticed your reflection on the concave and convex sides of a spoon? The smooth surface acts like a

B. Mirages

Mirages happen when the ground is very hot and the air is cool. The hot ground warms a layer of air just above

C. Haloes, sundogs, primary rainbows, secondary rainbows, and supernumerary bows

Haloes form when sunlight or moonlight is refracted by ice crystals in high, thin clouds

D. Why are clouds usually white and rain clouds dark?

Mie scattering makes clouds look white. Cloud droplets about 20 micrometers wide scatter all visible light

E. Why is the sky blue and sunsets are reddish?

Rayleigh scattering causes blue skies and red-orange sunrises and sunsets. It happens when tiny particles, like gas

Key terms

Supernumerary bows - faint, colorful bands inside the primary rainbow, caused by interference of light. They

15. Special theory of Relativity

I. Newtonian Mechanics Vs Maxwell's Electromagnetic Theory

In 1886, Heinrich Hertz experimentally proved the existence of electromagnetic waves, as predicted by James

II. Special Theory of Relativity

1. Postulates of Special Theory of Relativity

A. Postulate #1: Principle of Relativity

The laws of Physics are the same in all inertial frames of reference moving with constant velocity relative

B. Postulate #2: Constancy of the Speed of Light

The speed of light is the same in all inertial frames of reference

2. Consequences of Special Relativity

A. Time Dilation

In an inertial system moving at high speed, time is observed to pass slowly.

B. Length Contraction

Objects appear shorter in the direction of motion when viewed from a stationary frame.

C. Mass - Energy Equivalence

Mass and energy are two forms of the same thing.

16. Postulates of General Relativity

Key terms

Since space time - more curved in regions where the gravitational field is strong, we would expect light passing

17. Speeds and Distances of Far-off Objects

Key terms

Parallax - apparent change in the position of an object due to change in the way it is perceived, depending on
Cepheid stars - very abundant and bright in the galaxy. If somebody obtains the distance to a given galaxy

18. The Expanding Universe

1. The open universe that tends to expand without end

1. Cosmic Microwave Background Radiation (CMB)

Discovery: Detected in 1965 by Arno Penzias and Robert Wilson.

2. Expansion of the Universe

Explanation: Hubble observed that galaxies are moving away from us in all directions, with velocities proportional

3. Abundance of Light Elements

Discovery: Predicted by George Gamow, Ralph Alpher, and Robert Herman in the 1940s; confirmed through

4. Large-Scale Structure of the Universe

Explanation: The distribution of galaxies and galaxy clusters in the universe shows a web-like structure.

5. Redshift of Galaxies

Explanation: The light from distant galaxies is redshifted, meaning it has shifted to longer wavelengths.

6. Primordial Gravitational Waves (Theoretical)

Explanation: Predicted ripples in spacetime created during the inflationary period of the early universe.

Key terms

H - the Hubble's constant equal to =